OFFICIAL SUBMISSION: CHECKPOINT GAME JAM 2026



ECHO-ISM

Initial Game Design Document & Concept Paper

A Top-Down Temporal Consensus Puzzle Game Translating Violent Radicalization Pathways into Systemic Mechanics of Polarization, Misinformation, and Collaborative De-escalation.

Jam Organizer:	Global Game Jam (GGJ), UNOCT, UNICRI
Competition Track:	Checkpoint Game Jam (Indonesia, Malaysia, Philippines, Singapore)
Core Challenge Theme:	Recognize and counter pathways to radicalization to violence
Target Engine & Tech:	Godot Engine 4.7.2 .NET / C# (Native Linux & Windows x86_64)
Version & Date:	v1.0.0 Release Candidate | November 1, 2026

1. Executive Summary & High Concept

ECHO-ISM is a 2D top-down time-loop puzzle game built for the Checkpoint Game Jam. Players control **Greg**, a humble maintenance courier inside **NEXUS**, a miniature digital city connected by a central communication and energy network known as the **Heartline**.

When an anonymous, manipulated broadcast triggers mass paranoia and distrust, the districts of NEXUS begin disconnecting their relays, locking security gates, and stockpiling resources. An autonomous signal entity named **NULL** exploits this growing isolation to sever communication routes and push the city toward a cascading collapse.

Greg cannot save the city through combat or coercion. Instead, using his **Chrono Echo** ability, Greg records his actions into temporal loops. By coordinating with ghostly echoes of his past actions, Greg operates parallel systems, holds open defensive gates for isolated residents, verifies fragmented telemetry, and restores collective consensus before radicalization cascades destroy the community.

> **Core Gameplay Metaphor**
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> In an echo chamber, isolated voices repeat without external input, amplifying panic. In ECHO-ISM, the player subverts the echo chamber: past attempts are transformed from self-reinforcing loops into cooperative teammates who bridge divides and re-establish connection.

2. Thematic Alignment: Countering Radicalization Pathways

Radicalization to violence is not an instantaneous event; it is an iterative social and psychological process characterized by distinct phases. ECHO-ISM translates the established frameworks of radicalization pathways into diegetic game systems without lecturing or reproducing real-world hate symbols.

Radicalization Pathway Phase	Systemic In-Game Manifestation (NEXUS)	Player Mechanical Countermeasure
1. Perceived Threat & Grievance	NULL injects a falsified emergency alert claiming peripheral districts are stealing central power.	**Source Inspection:** Greg queries corrupted terminals to retrieve unadulterated log files, distinguishing manufactured grievances from reality.
2. Information Isolation (Echo Chambers)	Districts close blast gates and disable transceivers, shutting out neighboring viewpoints and data.	**Bridge-Holding Loops:** Greg sends an echo to hold open physical isolation gates while his live self navigates across separated sectors.
3. Out-Group Dehumanization	NPCs like BOLT retreat into defensive postures, viewing all outsiders as hostile saboteurs.	**Empathetic De-escalation:** Non-violent NPC dialogue mechanics require listening to resident anxieties rather than attacking defensive behaviors.
4. Cascade Crisis / Mobilization	Overloaded junctions and timed security overrides threaten permanent physical disruption across NEXUS.	**Synchronized Multi-Loop Consensus:** Solving multi-station rotary junctions and key sockets simultaneously through coordinated temporal echoes.

Pedagogical Philosophy

- **No Combat, No Villainizing:** The antagonist is not defeated by destruction. Residents who succumbed to panic are treated as people with incomplete information needing reconnection, not irredeemable hostiles.
- **Agency and Active Prevention:** Players practice anticipation and preventative timing rather than reactive punishment.
- **Abstracted Integrity:** By setting the conflict in a stylized digital society, the game avoids real-world sectarian or ethnic stereotyping while remaining universally relevant.

3. Narrative Architecture & Dramatis Personae

The narrative of ECHO-ISM unfolds across five connected industrial and computational sectors. The narrative is communicated through environmental state changes, terminal logs, and concise NPC interactions.

Character	Design	Role in Society	Narrative Arc & Pathway Alignment
Greg	Cyan blob with expressive eyes	Maintenance Courier	The player character. Represents grassroots civic action, empirical verification, and persistent problem solving.
Pip	Green agile blob	Sector Courier	Greg's companion. Represents the early witness who encounters contradictory information and risks falling into despair if isolated.
Mira	Yellow geometric archivist	Historical Record Keeper	Guards the immutable archive logs. Embodies evidence verification and institutional memory against revisionist disinformation.
BOLT	Orange defensive sentinel	Power Station Engineer	Initially hostile and paranoid. Illustrates how fear and perceived resource scarcity drive defensive extremism and preemptive aggression.
NULL	Violet glitch entity	Corrupted Routing Process	The systemic antagonist. Operates not by physical strength, but by injecting latency, distorting packet payloads, and polarizing feedback loops.

> **Narrative Resolution: The Consensus Protocol**
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> In the final sector (Sector 05: Heartline), Greg does not single-handedly flip a master switch. The ending requires Pip, Mira, BOLT, and Greg's temporal echoes to stabilize four distinct relays simultaneously, demonstrating that countering polarization requires community-wide consensus.

4. Core Mechanics & Interactive Puzzle Systems

ECHO-ISM is built around tight, deterministic top-down controls coupled with a temporal recording buffer.

4.1 The Chrono Echo Loop

- **Recording Phase:** Greg moves with responsive top-down kinematics (WASD / Arrow Keys). Every input, position delta, and interaction timestamp is stored in a circular ring buffer.
- **Loop Timer:** Each sector features a localized loop timer (10 to 30 seconds). Players can either let the timer expire or press `[R]` to manually trigger a temporal reset.
- **Playback Phase:** Upon reset, Greg's prior action path is spawned as a semi-transparent cyan ghost (an Echo). The Echo faithfully reproduces all movement, pressure-plate activation, and station interactions.
- **Multi-Echo Concurrency:** Later sectors permit up to 4 concurrent echoes, demanding precise temporal choreography where an echo in loop 1 unlocks a gate for an echo in loop 2.

4.2 Modular Puzzle Components

Module	Physical Behavior	Cooperative Puzzle Application
Pressure Plate	Depresses when stepped on; springs back when vacated.	Requires an echo to stand continuously on the plate to hold open blast doors for the live player.
Rotary Junction	Rotates power conduit channels 90 degrees per interaction.	Requires coordinating rotation angles across multiple junctions to route energy packets without interruption.
Timed Gate	Opens for a strict duration (e.g. 4.0 seconds) after activation.	Demands strict synchronization: an echo triggers the gate timer while the live player sprints to cross before closure.
Data Key & Socket	Physical cryptographic key tokens carried by Greg.	One echo retrieves and inserts the key into socket A, rerouting power to open a dispenser for socket B.
Charge Station	Requires continuous channeling (3.0 seconds) to prime.	Allows echoes to absorb the commitment time so the player can maneuver through downstream corridors.

5. Campaign Structure: 5 Connected Sectors

The campaign features five hand-crafted sectors, each mapping directly to a stage in the de-escalation of radicalized panic.

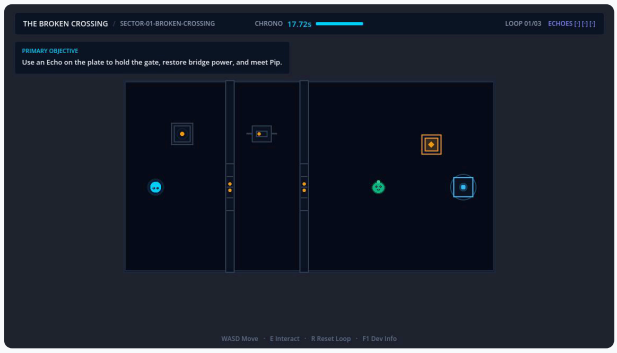


Figure 5.1: Sector 01 - The Broken Crossing (Bridging Initial Distrust)

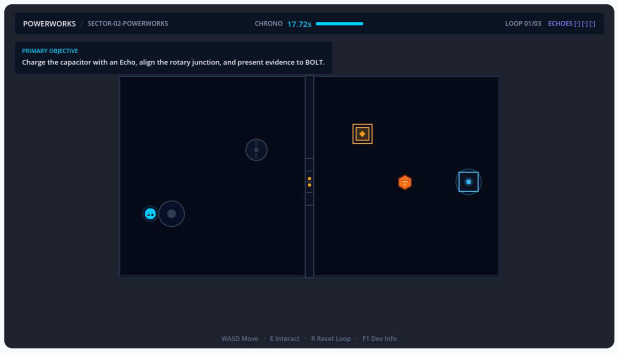


Figure 5.2: Sector 02 - Powerworks (De-escalating BOLT)

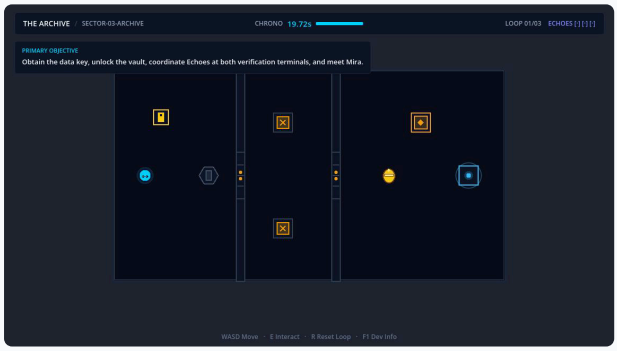


Figure 5.3: Sector 03 - The Archive (Empirical Fact Verification)

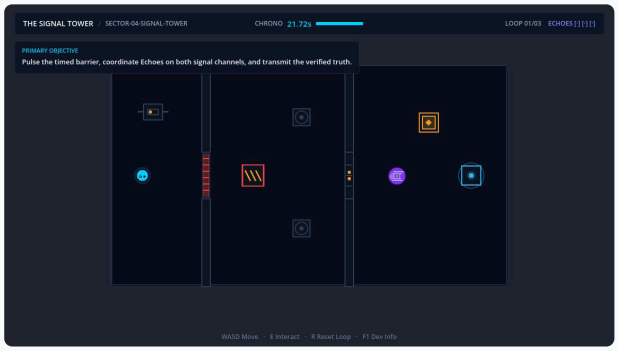


Figure 5.4: Sector 04 - The Signal Tower (Isolating Disinformation Vectors)

Sector & Title	Thematic Metaphor	Primary Puzzle Challenge
Sector 01: The Broken Crossing	Recognizing false rumors that divide neighbors.	Coordinate an echo on a remote pressure switch to lower security barriers and rescue Pip.
Sector 02: The Powerworks	De-escalating defensive paranoia and hoarding.	Bypass perimeter lockouts and interact peacefully with BOLT to restore municipal current.
Sector 03: The Archive	Recovering objective ground truth and context.	Collect fragmented cryptographic tokens across timed puzzle chambers to reconstruct original logs.
Sector 04: The Signal Tower	Silencing malicious disinformation amplifiers.	Simultaneously calibrate three rotary broadcast junctions before NULL's signal overrides the tower.
Sector 05: The Heartline	Building community resilience and consensus.	Multi-loop synchronization: four terminal locks must be sustained in parallel to re-link all five districts.

6. Accessibility, UX & Technical Implementation

6.1 Accessibility & Anti-Slop Visual Standards

- **WCAG AAA Contrast:** All interactive elements, text, and critical cues maintain a contrast ratio greater than 7:1 against their backgrounds.
- **Multi-Channel Feedback:** Puzzle states never rely solely on color. Active, locked, verified, and unverified states use distinct silhouettes, pulsing traces, and animated glyphs.
- **Comfortable Controls:** Support for keyboard (WASD / Arrows + Space / E) and gamepad. Zero rapid-fire QTEs; difficulty stems from logic and choreography, not twitch execution.

6.2 Engine & Architecture

- **Engine:** Godot Engine 4.7.2 .NET (C#) running natively on both Linux x86_64 and Windows x86_64.
- **Deterministic Physics:** Fixed tick physics processing (60 Hz) guarantees replay consistency across loops without desynchronization.
- **Automated Regression Testing:** Headless CI testing harness executing 216 automated assertions covering loop rewind, module state persistence, collision offsets, and victory conditions.

> **Submission Verification**
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> ECHO-ISM was developed exclusively for the Checkpoint Game Jam 2026. All source code, scenes, test suites, and promotional materials are version-controlled in the official repository.